

Sushan Adhikari

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RESEARCH INTERESTS

Low-resource NLP · Legal Machine Translation · Multilingual NLP · Explainable AI · Retrieval-Augmented Generation · LLM Alignment

EDUCATION

B.Eng. Computer Engineering — Kathmandu University, School of Engineering Jan 2022 – Feb 2026
CGPA: 3.90/4.0 [\[Transcript\]](#) Coursework: Data Structures & Algorithms, Computer Vision, Databases, Computer Architecture, Advanced Calculus, Statistics & Probability, Differential Equations

Exchange Semester, B.Tech CSE — Indian Institute of Technology Palakkad Jan – May 2025
Coursework: Introduction to AI, Linear Algebra, Compiler Design, Graph Theory & Combinatorics

RESEARCH & PUBLICATIONS

Conference

- **Directional Asymmetry in Low-Resource Legal Machine Translation: A Nepali–English Case Study**
Sushan Adhikari, S. Sharma, D. Lamichhane, R. Chulyadyo, B. K. Bal *ACM ICAIL 2026, Jun 2026* [\[DOI processing\]](#)
[\[manuscript\]](#) · [\[Acceptance Email\]](#) · [\[GitHub\]](#) · [\[Poster\]](#)

Journal

- **Enhancing Ethical Reasoning in Tiny LLMs via Fine-Tuning and Multi-Agent Consensus**
Sushan Adhikari, S. Sharma, D. Lamichhane, S. Sigdel *KUSET Vol.18 No.2, Dec 2024*
[\[manuscript\]](#) · [\[GitHub\]](#)
- **Detecting Image Forgeries and Deepfakes: A Comparative Study of CNN and Transformer Models**
P. Kafle, Sushan Adhikari, A. M. Shakya, N. Ghimire, G. Sharma *KUSET Vol.18 No.2, Dec 2024*
[\[manuscript\]](#)
- **Physics-Informed Data Augmentation for Sediment Concentration Prediction in Himalayan Rivers**
U. Adhikari, M. R. Kafle, Sushan Adhikari *InJET-InDev Vol.2 No.2, Feb 2026*
[\[manuscript\]](#) · [\[LEC Certificate\]](#)

Preprints

- **AlgoRAG: Retrieval-Augmented Generation for Theoretical Computer Science Education**
Sushan Adhikari *arXiv:2609.14572, Sep 2026*
[\[arXiv\]](#) · [\[GitHub\]](#)
- **How Much Data Is Enough? Validation Gaps in Manim-Based Mathematical Animation Code Generation**
Sushan Adhikari *Preprint, Sep 2026*
[\[preprint\]](#) · [\[GitHub\]](#)
- **Explainability Methods for Hybrid CNN-Transformer Document Layout Analysis**
Sushan Adhikari *Ongoing*
[\[preprint\]](#) · [\[GitHub\]](#)

RESEARCH EXPERIENCE

Low-Resource Legal Machine Translation & Corpus Construction 2025–2026
First-authored · ACM ICAIL 2026 · Kathmandu University

- Built the first public Nepali–English legal parallel corpus (5,024 pairs) from three authoritative sources via four-stage pipeline: Devanagari OCR (PyTesseract), rule-based clause-preserving segmentation, semi-automatic alignment with bilingual law-student verification, and lawyer quality control; 8% of alignments required re-segmentation.
- Designed 6-model experiment (mBART-50 and NLLB-200 × 3 configs); bidirectional joint training gained +4.92 BLEU Ne→En but lost 6.34 BLEU En→Ne on mBART-50 — pivoted paper framing to directional asymmetry after observing results; attributed divergence to morphological interference from Nepali's agglutinative structure; released corpus, models, and code publicly.

Ethical Reasoning in Tiny Language Models via Multi-Agent Consensus 2024
KUSET Vol.18 No.2, Dec 2024 · Kathmandu University

- Synthesized 1,000-scenario ethical dilemma dataset (Gemini 2.5 Pro) with structured QA (40% accept / 45% revise / 15% reject); LoRA fine-tuned three TinyLlama-1.1B agents specialized for utilitarian, deontological, and virtue-ethics frameworks; confidence-weighted softmax consensus aggregated outputs.
- Agents achieved philosophical consistency of 97.8 / 95.2 / 96.5; validated by 55-participant human study (multi-agent quality 94.2%); released dataset and code publicly.

Deepfake and Image Forgery Detection: CNN vs. Transformer Benchmark 2024
KUSET Vol.18 No.2, Dec 2024 · Kathmandu University (corresponding author)

- Curated a 140,000+ image dataset spanning CelebA/Flickr real images, classical manipulation (splicing, copy-move, face-swap, inpainting), and GAN-generated synthetic faces (StyleGAN, StyleGAN2) — first benchmark combining both classical and generative forensic categories.
- InceptionV3 (22.8M params) outperformed ViT-Base (86.3M params) by 5.51% accuracy and 10.8% recall (AUC-ROC 0.986 vs. 0.884); confirmed CNN advantage over transformers on limited forensic datasets.

Physics-Informed Data Augmentation for Himalayan Sediment Prediction

2025–2026

InJET-InDev Vol.2 No.2, Feb 2026 · IOE Tribhuvan University & Kathmandu University

- Wrote all ML and data engineering code for a 51-year Budhigandaki River record (560 valid monthly observations); benchmarked 10 augmentation strategies — 5 classical and 5 physics-informed (seasonal stochastic rating curves, k-NN discharge analogs, STL decomposition, physics-constrained ensemble) — under 5-fold CV.
- Physics-informed ensemble outperformed best classical baseline: R^2 +5.5%, RMSE –24.3%, MAE –47.6%; three-method consensus outlier detection preserved 99.8% of observations.

AlgoRAG: Domain-Specialized RAG for Theoretical CS Education

2025–2026

arXiv:2609.14572 · Kathmandu University

- Built a five-stage RAG pipeline over 1,200+ TCS artifacts (847 lecture slides, 156 proof templates, 312 practice problems): SciBERT-based mathematical entity recognition, hybrid dense (Sentence-BERT) + sparse retrieval over ChromaDB with a CS knowledge graph, and pedagogical re-ranker scoring on explanation clarity and query-difficulty alignment.
- BLEU-4 = 0 across all 179 exam queries (logically equivalent proofs differ lexically); proposed 6-criterion rubric-based pedagogical quality metric as domain-appropriate alternative; achieved 100% response success and 0.7620 pedagogical quality.

Manim Code Generation: Dataset Scaling and Execution-Based Validation

2026

Preprint, Sep 2026 · Kathmandu University

- Constructed ManimCalc-1493 — 1,493 execution-validated Manim calculus animation scripts across 20 function categories ($3.6\times$ prior largest corpus); four-stage validation: syntax → AST parse → Manim dry-run execution → API hallucination detection.
- QLoRA fine-tuned (4-bit NF4, rank 64) DeepSeek-Coder-1.3B/6.7B and Phi-2 (87–92% training-loss reduction); proposed 3-stage weak→AST→execution evaluation framework to quantify the validation gap; zero-shot baseline: 0% execution success confirming gap without fine-tuning.

XAI for Hybrid CNN-Transformer Document Layout Analysis

2026

Ongoing Preprint · Kathmandu University

- Trained ResNet-50 + Deformable Transformer hybrid on 20,000 PubLayNet samples (val loss 1.4384); compared Grad-CAM, Attention Rollout, and LIME on 50 documents; introduced 4 novel metrics: Boundary Alignment Score, Structural Consistency, Element-Specific Fidelity, Human Interpretability Rating.
- LIME achieved highest boundary alignment (BAS = 0.325); Attention most consistent (SC = 0.539 ± 0.020); confirmed complementarity: CNN localizes edges, Transformer captures global page structure.

TECHNICAL SKILLS

Languages & ML: Python; PyTorch, Scikit-Learn, OpenCV, YOLOv8; Pandas, NumPy, HuggingFace
Inference & CV: NVIDIA DeepStream, Triton Inference Server, GStreamer, TensorRT
MLOps & Backend: Docker, AWS, Git, FastAPI, PostgreSQL, Redis
Other: ChromaDB, LangChain, L^AT_EX/Overleaf

WORK EXPERIENCE

- **Nurvexa** — Co-founder (startup) nurvexalabs.com Jul 2025 – Present
AI medical licensing exam prep for Nepal's healthcare ecosystem; multi-stage RAG pipeline with SSE token streaming, 2,200+ OCR-extracted quiz questions, Gemini Vision, hybrid search, cross-encoder rerankers, Redis, PostgreSQL, Docker, full-stack deployment.
- **Junior MLOps Developer** — KyraWorks Jan – Apr 2026 [\[Certificate\]](#)
Promoted from CV Intern; monorepo + CODEOWNERS governance across 10+ team members; SAST/DAST (DefectDojo, CodeRabbit); Docker; 3+ AWS EC2 instances via Terraform.
- **Computer Vision Intern** — KyraWorks Oct 2025 – Jan 2026 [\[Certificate\]](#)
Real-time inference with NVIDIA DeepStream and Triton; deployed YOLOv8 across 10+ simultaneous camera streams at ~60 fps aggregate throughput.

PROJECTS

- **Statistics & Probability for Machine Learning** *In Progress – updated daily* | [GitHub](#)
Public compendium: probability theory, MLE/MAP, Bayesian inference, Gaussian processes, information theory, and concentration inequalities with NumPy implementations.
- **Agni Netra: Forest Fire Detection System** 2024 | [GitHub](#)
Fine-tuned YOLOv8n for real-time fire/smoke detection on a Raspberry Pi; integrated ESP32 with GPS/GPRS for location-tagged alerts — built in response to Nepal's 6,279 recorded forest fires in 2021.

ACHIEVEMENTS & LEADERSHIP

- **Hult Prize OnCampus Winner** (2024) — CV-based fish-disease detection; represented Nepal at Regional Summit, Bangkok, Thailand.
- **Winner, Health Track** — AI Crusade 2023 (first AI hackathon in Nepal). [\[Certificate\]](#)
- **Mahatma Gandhi Scholarship** — Indian Embassy, Nepal (2019–2021).
- **Internal Relations & Ops Lead** — KU HackFest 2024 (48-hour hackathon, ~200 team applications).

CERTIFICATIONS

[Machine Learning Specialization](#) — Andrew Ng (Stanford/Coursera, 2026) · [Embeddings for NLP](#) (OpenHPI, 2025)

LANGUAGES

Nepali (Native) **English** C1 Advanced — [IELTS 8.0](#)